NETFLIX DATA ANALYSIS

(REPORT USING POSTGRESQL & POWER BI)

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1. INTRODUCTION

This project presents an analysis of Netflix movies and TV shows using PostgreSQL and Power BI.

The main goal of the project is to analyze Netflix content and extract meaningful insights from the dataset. The project includes data cleaning, handling missing values, duplicate detection, ranking analysis, and data visualization.

The system allows the user to:

- Analyze Movies vs TV Shows
- Analyze content production by year
- Analyze content distribution by country
- Discover top genres on Netflix
- Detect missing values and duplicates
- Use SQL analytical functions such as ROW_NUMBER() and RANK()
- Visualize the data using Power BI dashboards
- Automate the analysis using SQL scripts and Task Scheduler

The project helps in understanding Netflix content trends and provides visual insights using interactive dashboards.

2. DATA CLEANING

Idea:

Handle missing values in important columns before performing analysis.

```sql
-- ***** Handle Missing Values *****
UPDATE netflix
SET country = 'Unknown'
WHERE country IS NULL;

UPDATE netflix
SET "cast" = 'Unknown'
WHERE "cast" IS NULL;

UPDATE netflix
SET rating = 'Unknown'
WHERE rating IS NULL;

UPDATE netflix
SET duration = 'Unknown'
WHERE duration IS NULL;
```

- NULL values in important columns were replaced with the value “Unknown”.
- This prevents problems during filtering and visualization.
- Data cleaning improves analysis accuracy and dashboard consistency.

3. MOVIES VS TV SHOWS ANALYSIS

Idea:

Compare the number of Movies and TV Shows available on Netflix.

```sql
-- ***** Movies vs TV Shows Count *****
SELECT type,COUNT(*) AS total_count
FROM netflix
GROUP BY type;
```

- GROUP BY was used to separate data by content type.
- COUNT() calculates the total number of records for each type.

Expected Output:

- Total number of Movies
- Total number of TV Shows

Data Output Messages Notifications

	type text	total_count bigint
1	Movie	6131
2	TV Sho...	2578

4. ANALYSIS BY RELEASE YEAR

Idea:

Analyze Netflix content growth over the years.

```sql
-- ***** Analysis by Year *****
SELECT
    release_year,
    COUNT(*) AS total_titles
FROM netflix
GROUP BY release_year
ORDER BY release_year;
```

- The query groups content by release year.
- COUNT() calculates the number of titles released each year.
- ORDER BY sorts the results chronologically.

Expected Output:

A yearly trend showing growth in Netflix content production.

	release_year bigint	total_titles bigint
70	2017	1016
71	2018	1140
72	2019	1030
73	2020	953

5. ANALYSIS BY COUNTRY

Idea:

Analyze which countries produce the most Netflix content.

```sql
-- ***** Analysis by Country *****
SELECT
    TRIM(unnest(string_to_array(country, ',''))) AS country_name,
    COUNT(*) AS total_titles
FROM netflix
WHERE country IS NOT NULL
GROUP BY country_name
ORDER BY total_titles DESC;
```

- string_to_array() splits countries separated by commas.
- unnest() converts the array into rows.
- COUNT() calculates the number of titles per country.

Expected Output:

Countries ranked by total Netflix titles.

Data Output Messages Notifications

Showing rows: 1 to 124 Page No: 1

	country_name text	total_titles bigint
1	United States	3643
2	India	1045
3	Unknown	827
4	United Kingdom	787
5	Canada	432
6	France	389
7	Japan	314
8	Spain	228
9	South Korea	226
10	Germany	225
11	Mexico	169
12	China	162
13	Australia	157

6. TOP GENRES ANALYSIS

Idea:

Identify the most popular genres on Netflix.

```sql
-- ***** Top Genres *****
SELECT
    TRIM(unnest(string_to_array(listed_in, ','))) AS genre,
    COUNT(*) AS total
FROM netflix
GROUP BY genre
ORDER BY total DESC
LIMIT 10;
```

- Genres were separated using string functions.
- COUNT() calculates the frequency of each genre.
- LIMIT 10 displays the top genres only.

Expected Output:

Top 10 most common genres on Netflix.

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	genre text	total bigint
1	International Movies	2752
2	Dramas	2427
3	Comedies	1674
4	International TV Shows	1328
5	Documentaries	869
6	Action & Adventure	859
7	Independent Movies	756
8	TV Dramas	739
9	Children & Family Movi...	641
10	Romantic Movies	616

7. MISSING VALUES & DUPLICATES

Missing Values Count

```sql
-- ***** Missing Values Count *****
SELECT
    COUNT(*) FILTER (WHERE director IS NULL) AS missing_director,
    COUNT(*) FILTER (WHERE country IS NULL) AS missing_country,
    COUNT(*) FILTER (WHERE date_added IS NULL) AS missing_date,
    COUNT(*) FILTER (WHERE rating IS NULL) AS missing_rating,
    COUNT(*) FILTER (WHERE duration IS NULL) AS missing_duration
FROM netflix;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	missing_director bigint	missing_country bigint	missing_date bigint	missing_rating bigint	missing_duration bigint
1	0	0	0	0	0

Duplicates Detection

```sql
-- ***** Duplicates Count *****
SELECT
    title,
    type,
    COUNT(*)
FROM netflix
GROUP BY title, type
HAVING COUNT(*) > 1;
```

- Missing values were detected using FILTER().
- Duplicate records were detected using GROUP BY and HAVING.
- Due to already cleaned data set, outputs are zeroes in duplicates detected

8. USING ROW_NUMBER() AND RANK()

Idea:

Use analytical SQL functions for ranking and advanced analysis.

```sql
-- ***** Remove Duplicates (using ROW_NUMBER) *****
DELETE FROM netflix
WHERE show_id IN (
    SELECT show_id FROM (
        SELECT
            show_id,
            ROW_NUMBER() OVER (
                PARTITION BY title, type
                ORDER BY show_id
            ) AS rn
        FROM netflix
    ) t
    WHERE t.rn > 1
);
```

- Duplicate records were identified using GROUP BY and COUNT().
- ROW_NUMBER() was used to assign numbers to duplicate rows.
- Rows with rn > 1 were removed to keep only one unique record.

```sql

-- ***** Ranking genres by popularity *****
SELECT
    genre,
    total_titles,
    RANK() OVER (ORDER BY total_titles DESC) AS genre_rank
FROM (
    SELECT
        TRIM(unnest(string_to_array(listed_in, ',''))) AS genre,
        COUNT(*) AS total_titles
    FROM netflix
    GROUP BY genre
) t;
```

- Genres were ranked based on the number of titles available.
- RANK() assigns the same rank to genres with equal counts.
- This helps identify the most popular Netflix genres.

Data Output Messages Notifications

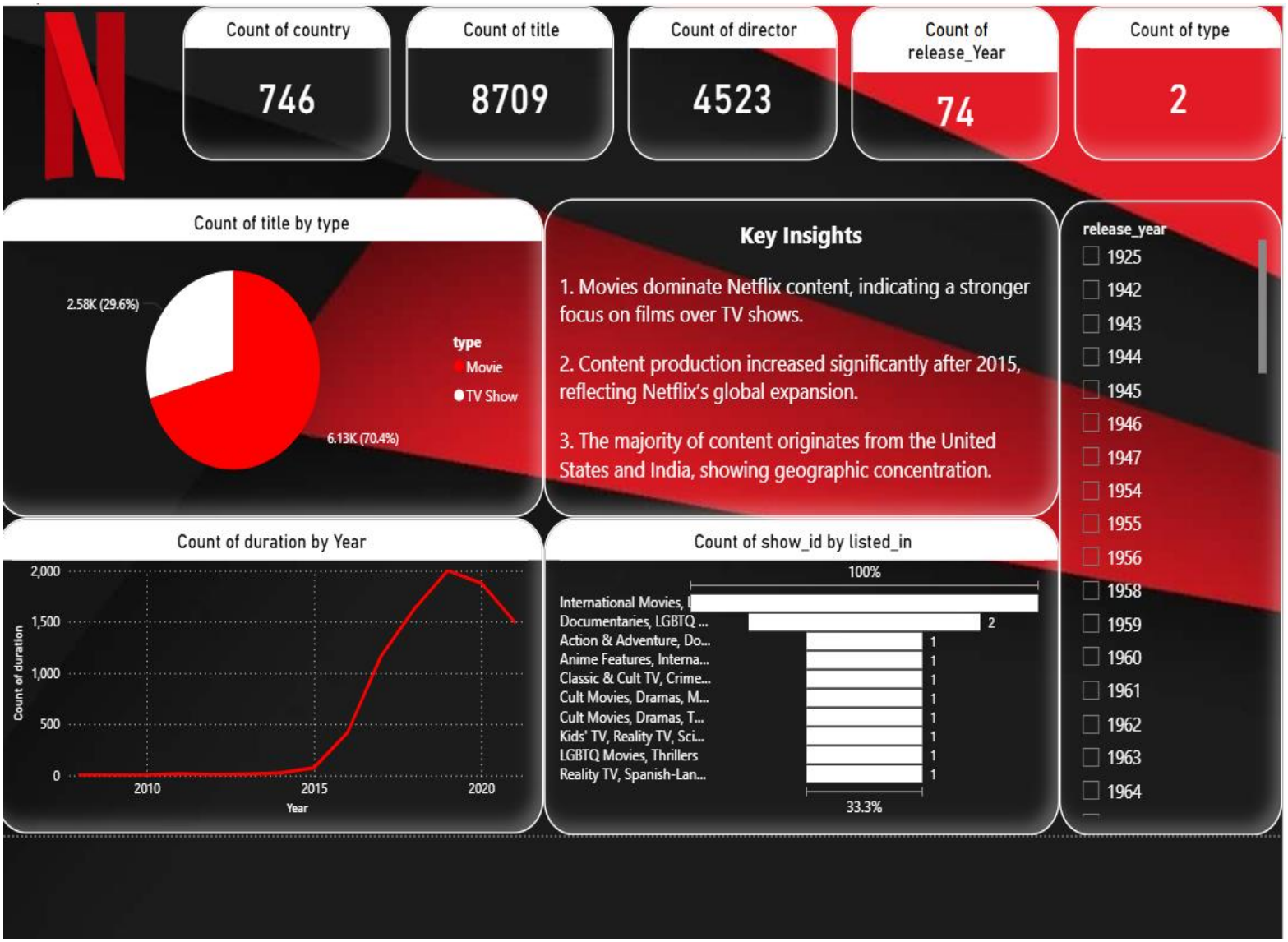
Showing rows: 1 to 42 Page No: 1

	genre text	total_titles bigint	genre_rank bigint
1	International Movies	2752	1
2	Dramas	2427	2
3	Comedies	1674	3
4	International TV Shows	1328	4
5	Documentaries	869	5
6	Action & Adventure	859	6

9. DASHBOARD VISUALIZATION

Power BI was used to create an interactive and colorful dashboard for visualizing Netflix data analysis results.

The dashboard page was designed with an organized modern layout that combines KPI cards, charts, slicers, and interactive visuals to improve data exploration and user experience.



Dashboard Components

1. KPI Cards

Several cards were added at the top of the dashboard to display important statistics such as:

- Count of Countries
- Count of Release Years
- Count of Directors
- Count of Titles
- Count of Content Types

These cards provide a quick overview of the dataset and help users understand the scale of Netflix content instantly.

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2. Pie Chart

A Pie Chart was used to compare:

- Movies
- TV Shows

This visualization clearly shows the distribution of Netflix content types and highlights which type dominates the platform.

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3. Line Chart

A Line Chart was created to visualize:

- Number of titles released over the years

This chart helps identify trends and growth patterns in Netflix content production across different years.

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4. Funnel Chart

A Funnel Chart was used to display ranked content categories and highlight the most dominant values in the dataset.

The funnel visualization helps simplify comparisons between large and small categories in an attractive way.

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5. Slicer

An interactive slicer was added to allow filtering dashboard results dynamically.

The slicer improves user interaction by enabling focused analysis based on selected values.

10. KEY INSIGHTS

1. Movies represent the majority of Netflix content compared to TV Shows.
2. Netflix content production increased significantly after 2015, indicating rapid platform growth.
3. The United States and India are among the leading countries producing Netflix content.

11. AUTOMATION USING SCRIPT & TASK SCHEDULER

Idea:

Automate the execution of the analysis script.

Implementation:

- The SQL queries were combined into a single SQL script file.
- A batch (.bat) file was created to run the script automatically.

- Windows Task Scheduler was used to schedule execution.

Benefits:

- Automation reduces manual work.
- Ensures regular execution of analysis.
- Makes the project more scalable and professional.

CONCLUSION

This project successfully analyzed Netflix data using PostgreSQL and Power BI. Different SQL techniques were used including aggregation, filtering, analytical functions, and data cleaning.

The dashboard provided interactive visual insights into Netflix content trends, genres, countries, and yearly growth.

The project demonstrates practical skills in SQL analysis, database management, and data visualization.